WEEK 2 · LESSON 11 OF 24 · HANDOUT

Reinforcement Learning

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=T_X4XFwKX8k — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

Your goal in this lesson: Describe how reinforcement learning works using agent, environment, action and reward, explain the explore-exploit trade-off, and say how it helps train chatbots.

Learning by trial and reward

Reinforcement learning (RL) is how an agent learns by **doing**. There are no labelled answers. Instead the agent tries actions, sees what happens, and receives **rewards** (good) or **penalties** (bad). Over many tries it learns which actions lead to the most reward over time.

Think of teaching a dog a new trick, or how you learned to ride a bicycle: nobody gave you a table of correct answers — you tried, fell, adjusted and improved.

The RL loop

1. The **agent** observes the **state** of the environment.
2. It chooses an **action**.
3. The environment changes to a new state and gives a **reward**.
4. The agent updates its **policy** — its strategy for choosing actions.
5. Repeat, thousands or millions of times.

Term	Meaning	Example: a robot vacuum
Agent	The learner	The robot
Environment	The world it acts in	The room
State	The current situation	Position, battery level, dirt nearby
Action	A choice	Move forward, turn, return to charger
Reward	Feedback signal	+1 for cleaning dirt, −10 for hitting stairs
Policy	The learned strategy	"When battery is low, head to the charger"

Long-term thinking

The agent aims for the **highest total reward over time**, not just the next reward. A chess move that loses a piece now may win the game later. RL handles **sequential** problems, where today's action changes tomorrow's options.

Explore or exploit?

Should the agent use what already works (**exploit**) or try something new that might be better (**explore**)? A shop owner who always stocks the same products never learns that a new product might sell better; one who changes everything every week cannot build on success. Good RL agents balance both — exploring more at the start, exploiting more as they learn.

Reward design is everything

The agent will chase whatever is rewarded — even in ways nobody intended. If a cleaning robot is rewarded for "dust collected", it might learn to spill dust and collect it again. This is called **reward hacking**, and it connects to the PEAS lesson: **what you measure is what you get.**

Where RL is used

- **Games** — AlphaGo learned partly by playing millions of games against itself.
- **Robotics** — learning to walk, grasp or balance.
- **Resource control** — managing energy use in data centres, traffic signals.
- **Chatbots — reinforcement learning from human feedback (RLHF)**: people rank several model answers from best to worst, a reward model learns those preferences, and the chatbot is tuned to give answers people prefer. This is one reason modern assistants are more helpful and polite than raw language models.

The three types of machine learning

	Supervised	Unsupervised	Reinforcement
Learns from	Labelled examples	Unlabelled data	Rewards from actions
Question	"What is the answer?"	"What structure is here?"	"What should I do next?"

Remember this: Reinforcement learning learns by trial, error and reward. It suits sequential decisions, depends on careful reward design, and helps make chatbots more useful through RLHF.

Name: _____ Date: _____

Activity

Task 1

Design an RL setup for a traffic light at a busy Kigali junction: write the state, actions and reward.

Task 2

Describe one way the agent could "hack" your reward, and fix the reward to prevent it.

Task 3

Give one example from your own life of exploring versus exploiting.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. How does a reinforcement-learning agent learn?
   - A. By copying a website
   - B. From a table of correct answers
   - C. By trying actions and receiving rewards or penalties
   - D. By clustering data
2. What is a 'policy' in RL?
   - A. The training data
   - B. A government law
   - C. The reward value
   - D. The agent's learned strategy for choosing actions

3. Why might an agent sacrifice a small reward now?

A. It aims for the highest total reward over time
B. It is broken
C. To save electricity
D. Rewards do not matter

4. What is the explore-exploit trade-off?

A. Choosing between two computers
B. Choosing between labels
C. Choosing between trying new actions and using actions known to work
D. Choosing a programming language

5. A cleaning robot spills dust so it can collect it again for more reward. This is…

A. Good design
B. Reward hacking
C. Clustering
D. Data cleaning

6. In RLHF, what do people provide?

A. New computer chips
B. Labels for every word
C. Rankings of model answers, which teach a reward model their preferences
D. Electricity

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